SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Concept Mapping
Annexure A

Criteria	Concept Identification ***(2.5)***	Linkages ***(2.5)***	Organization ***(2)***	Integration of Literature ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I A. LOKESH KUMAR(192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A Oki

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Annexure A

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

🎯 What your map must show:

1. Core components: Data, Model, User Interface
2. Types of DSS
3. Flow: Data → Analysis → Decision
4. Role in business decision-making
5. Real-world example linkage

Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts

2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

🎯 What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME)

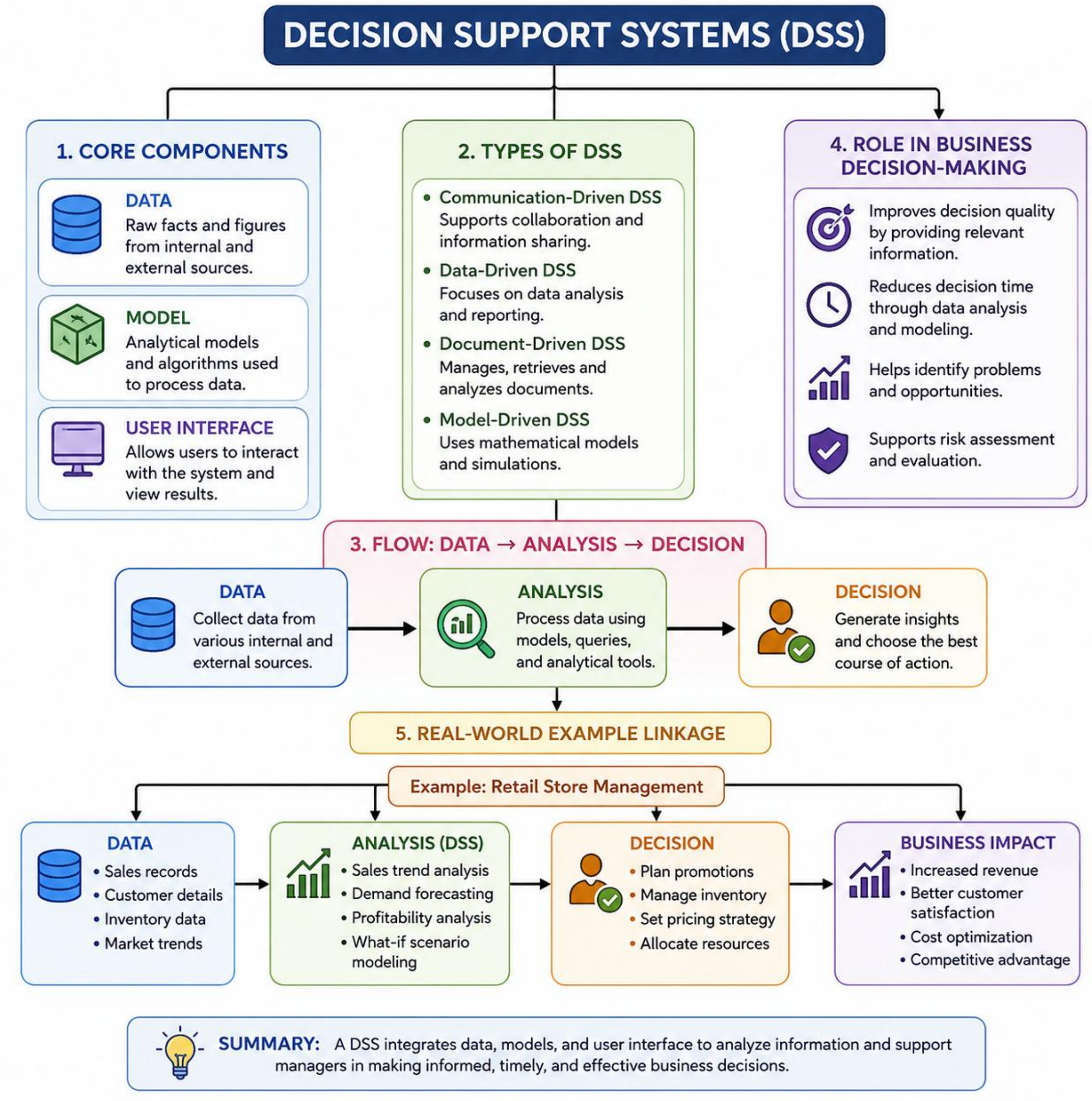
Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:

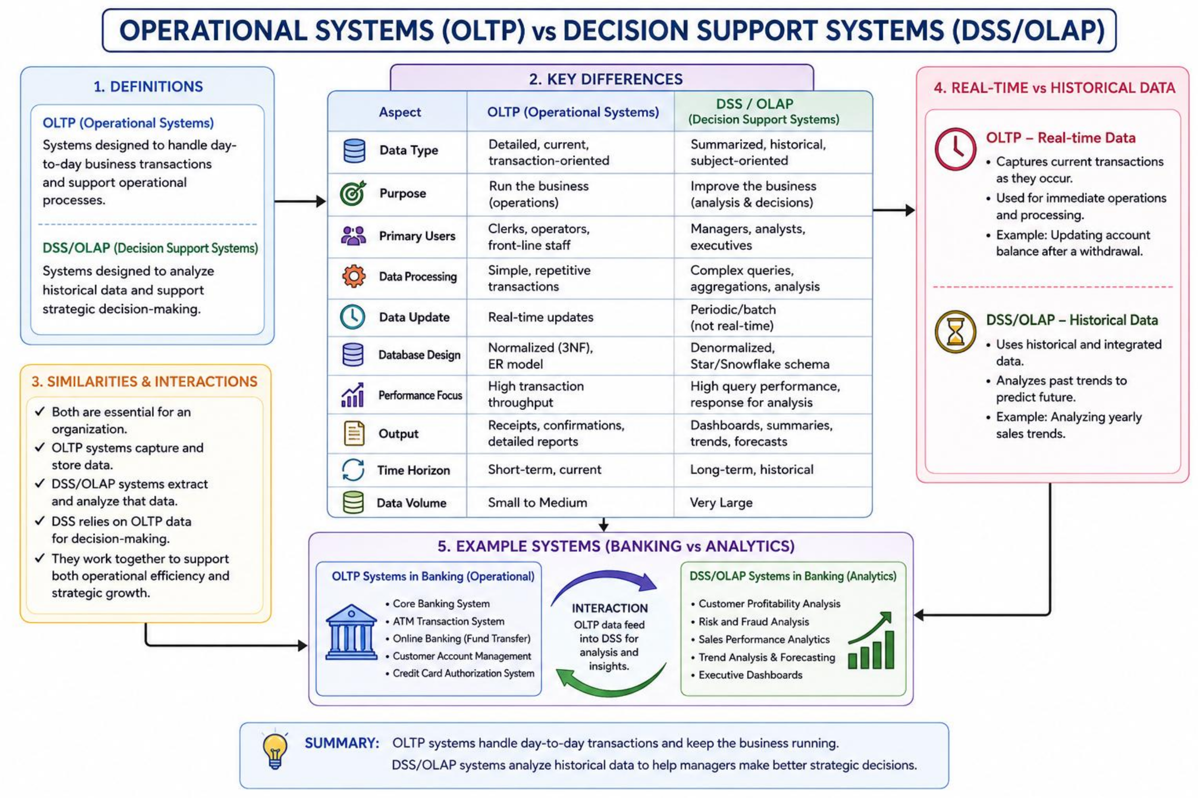
1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integration

1.



DECISION SUPPORT SYSTEMS (DSS)
1. CORE COMPONENTS
DATA
Raw facts and figures from internal and external sources.
MODEL
Analytical models and algorithms used to process data.
USER INTERFACE
Allows users to interact with the system and view results.
2. TYPES OF DSS
• Communication-Driven DSS
Supports collaboration and information sharing.
• Data-Driven DSS
Focuses on data analysis and reporting.
• Document-Driven DSS
Manages, retrieves and analyzes documents.
• Model-Driven DSS
Uses mathematical models and simulations.
4. ROLE IN BUSINESS DECISION-MAKING
Improves decision quality by providing relevant information.
Reduces decision time through data analysis and modeling.
Helps identify problems and opportunities.
Supports risk assessment and evaluation.
3. FLOW: DATA → ANALYSIS → DECISION
DATA
Collect data from various internal and external sources.
ANALYSIS
Process data using models, queries, and analytical tools.
DECISION
Generate insights and choose the best course of action.
5. REAL-WORLD EXAMPLE LINKAGE
Example: Retail Store Management
DATA
• Sales records
• Customer details
• Inventory data
• Market trends
ANALYSIS (DSS)
• Sales trend analysis
• Demand forecasting
• Profitability analysis
• What-if scenario modeling
DECISION
• Plan promotions
• Manage inventory
• Set pricing strategy
• Allocate resources
BUSINESS IMPACT
• Increased revenue
• Better customer satisfaction
• Cost optimization
• Competitive advantage
SUMMARY: A DSS integrates data, models, and user interface to analyze information and support managers in making informed, timely, and effective business decisions.

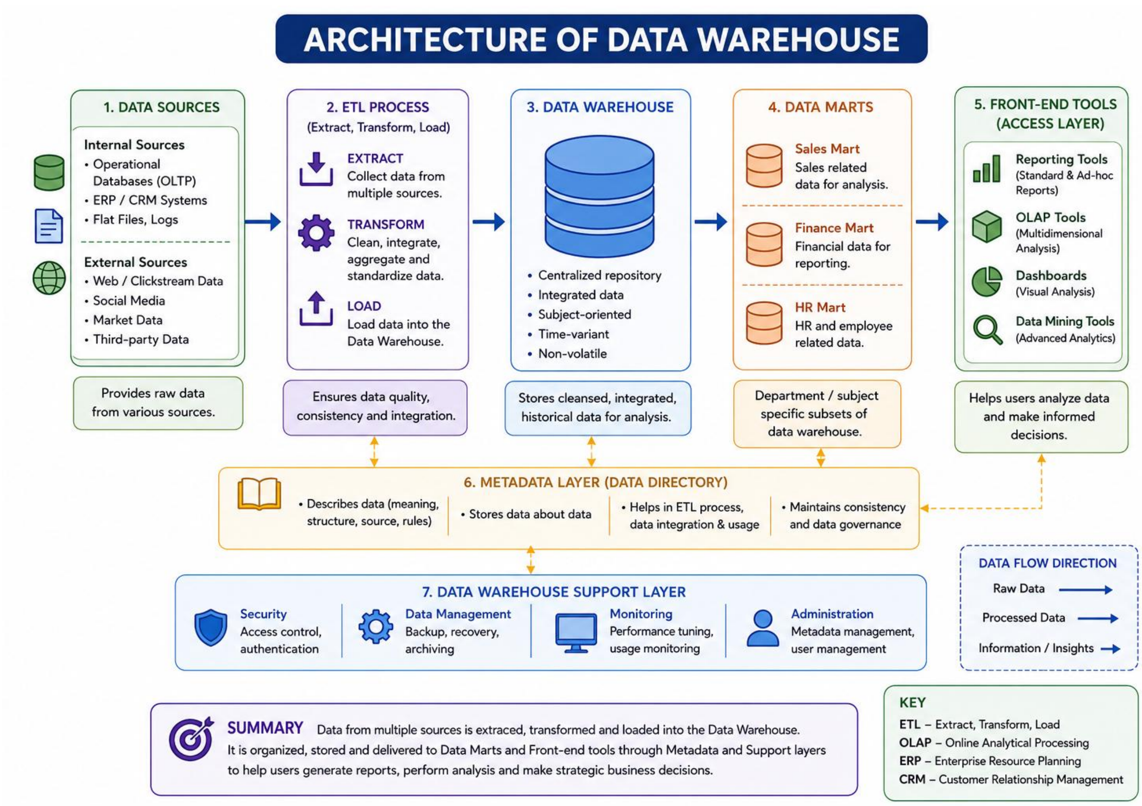
2.


OPERATIONAL SYSTEMS (OLTP) vs DECISION SUPPORT SYSTEMS (DSS/OLAP)
1. DEFINITIONS
OLTP (Operational Systems)
Systems designed to handle day-to-day business transactions and support operational processes.
DSS/OLAP (Decision Support Systems)
Systems designed to analyze historical data and support strategic decision-making.
2. KEY DIFFERENCES

Aspect	OLTP (Operational Systems)	DSS / OLAP (Decision Support Systems)
Data Type	Detailed, current, transaction-oriented	Summarized, historical, subject-oriented
Purpose	Run the business (operations)	Improve the business (analysis & decisions)
Primary Users	Clerks, operators, front-line staff	Managers, analysts, executives
Data Processing	Simple, repetitive transactions	Complex queries, aggregations, analysis
Data Update	Real-time updates	Periodic/batch (not real-time)
Database Design	Normalized (3NF), ER model	Denormalized, Star/Snowflake schema
Performance Focus	High transaction throughput	High query performance, response for analysis
Output	Receipts, confirmations, detailed reports	Dashboards, summaries, trends, forecasts
Time Horizon	Short-term, current	Long-term, historical
Data Volume	Small to Medium	Very Large

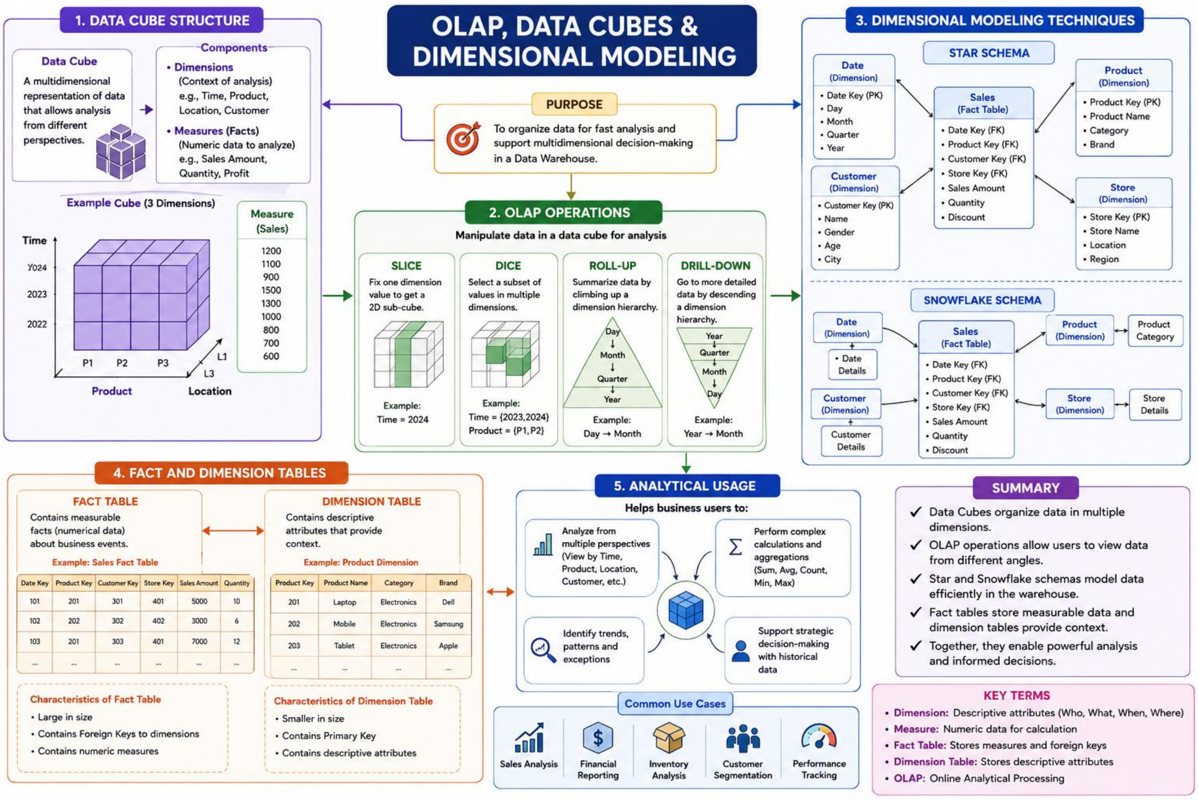
3. SIMILARITIES & INTERACTIONS
✓ Both are essential for an organization.
✓ OLTP systems capture and store data.
✓ DSS/OLAP systems extract and analyze that data.
✓ DSS relies on OLTP data for decision-making.
✓ They work together to support both operational efficiency and strategic growth.
4. REAL-TIME vs HISTORICAL DATA
OLTP – Real-time Data
• Captures current transactions as they occur.
• Used for immediate operations and processing.
• Example: Updating account balance after a withdrawal.
DSS/OLAP – Historical Data
• Uses historical and integrated data.
• Analyzes past trends to predict future.
• Example: Analyzing yearly sales trends.
5. EXAMPLE SYSTEMS (BANKING vs ANALYTICS)
OLTP Systems in Banking (Operational)
• Core Banking System
• ATM Transaction System
• Online Banking (Fund Transfer)
• Customer Account Management
• Credit Card Authorization System
INTERACTION
OLTP data feed into DSS for analysis and insights.
DSS/OLAP Systems in Banking (Analytics)
• Customer Profitability Analysis
• Risk and Fraud Analysis
• Sales Performance Analytics
• Trend Analysis & Forecasting
• Executive Dashboards
SUMMARY: OLTP systems handle day-to-day transactions and keep the business running.
DSS/OLAP systems analyze historical data to help managers make better strategic decisions.

3.



ARCHITECTURE OF DATA WAREHOUSE
1. DATA SOURCES
Internal Sources
• Operational Databases (OLTP)
• ERP / CRM Systems
• Flat Files, Logs
External Sources
• Web / Clickstream Data
• Social Media
• Market Data
• Third-party Data
Provides raw data from various sources.
2. ETL PROCESS
(Extract, Transform, Load)
EXTRACT
Collect data from multiple sources.
TRANSFORM
Clean, integrate, aggregate and standardize data.
LOAD
Load data into the Data Warehouse.
Ensures data quality, consistency and integration.
3. DATA WAREHOUSE
• Centralized repository
• Integrated data
• Subject-oriented
• Time-variant
• Non-volatile
Stores cleansed, integrated, historical data for analysis.
4. DATA MARTS
Sales Mart
Sales related data for analysis.
Finance Mart
Financial data for reporting.
HR Mart
HR and employee related data.
Department / subject specific subsets of data warehouse.
5. FRONT-END TOOLS (ACCESS LAYER)
Reporting Tools
(Standard & Ad-hoc Reports)
OLAP Tools
(Multidimensional Analysis)
Dashboards
(Visual Analysis)
Data Mining Tools
(Advanced Analytics)
Helps users analyze data and make informed decisions.
6. METADATA LAYER (DATA DIRECTORY)
• Describes data (meaning, structure, source, rules)
• Stores data about data
• Helps in ETL process, data integration & usage
• Maintains consistency and data governance
7. DATA WAREHOUSE SUPPORT LAYER
Security
Access control, authentication
Data Management
Backup, recovery, archiving
Monitoring
Performance tuning, usage monitoring
Administration
Metadata management, user management
DATA FLOW DIRECTION
Raw Data
Processed Data
Information / Insights
SUMMARY Data from multiple sources is extraced, transformed and loaded into the Data Warehouse.
It is organized, stored and delivered to Data Marts and Front-end tools through Metadata and Support layers
to help users generate reports, perform analysis and make strategic business decisions.
KEY
ETL – Extract, Transform, Load
OLAP – Online Analytical Processing
ERP – Enterprise Resource Planning
CRM – Customer Relationship Management

4.


OLAP, DATA CUBES & DIMENSIONAL MODELING
PURPOSE
To organize data for fast analysis and support multidimensional decision-making in a Data Warehouse.
1. DATA CUBE STRUCTURE
Data Cube
A multidimensional representation of data that allows analysis from different perspectives.
Components
• Dimensions (Context of analysis) e.g., Time, Product, Location, Customer
• Measures (Facts) (Numeric data to analyze) e.g., Sales Amount, Quantity, Profit
Example Cube (3 Dimensions)
Time
2024
2023
2022
P1
P2
P3
L1
L3
Product
Location
Measure (Sales)
1200
1100
900
1500
1300
1000
800
700
600
2. OLAP OPERATIONS
Manipulate data in a data cube for analysis
SLICE
Fix one dimension value to get a 2D sub-cube.
Example: Time = 2024
DICE
Select a subset of values in multiple dimensions.
Example: Time = {2023,2024} Product = {P1,P2}
ROLL-UP
Summarize data by climbing up a dimension hierarchy.
Day → Month → Quarter → Year
Example: Day → Month
DRILL-DOWN
Go to more detailed data by descending a dimension hierarchy.
Year → Quarter → Month → Day
Example: Year → Month
3. DIMENSIONAL MODELING TECHNIQUES
STAR SCHEMA
Date (Dimension)
• Date Key (PK) • Day • Month • Quarter • Year
Sales (Fact Table)
• Date Key (FK) • Product Key (FK) • Customer Key (FK) • Store Key (FK) • Sales Amount • Quantity • Discount
Product (Dimension)
• Product Key (PK) • Product Name • Category • Brand
Customer (Dimension)
• Customer Key (PK) • Name • Gender • Age • City
Store (Dimension)
• Store Key (PK) • Store Name • Location • Region
SNOWFLAKE SCHEMA
Date (Dimension)
• Date Details
Sales (Fact Table)
• Date Key (FK) • Product Key (FK) • Customer Key (FK) • Store Key (FK) • Sales Amount • Quantity • Discount
Product (Dimension)
Product Category
Customer (Dimension)
Customer Details
Store (Dimension)
Store Details
4. FACT AND DIMENSION TABLES
FACT TABLE
Contains measurable facts (numerical data) about business events.
Example: Sales Fact Table
Date Key | Product Key | Customer Key | Store Key | Sales Amount | Quantity
101 | 201 | 301 | 401 | 5000 | 10
102 | 202 | 302 | 402 | 3000 | 6
103 | 201 | 303 | 401 | 7000 | 12
... | ... | ... | ... | ... | ...
DIMENSION TABLE
Contains descriptive attributes that provide context.
Example: Product Dimension
Product Key | Product Name | Category | Brand
201 | Laptop | Electronics | Dell
202 | Mobile | Electronics | Samsung
203 | Tablet | Electronics | Apple
... | ... | ... | ...
Characteristics of Fact Table
• Large in size
• Contains Foreign Keys to dimensions
• Contains numeric measures
Characteristics of Dimension Table
• Smaller in size
• Contains Primary Key
• Contains descriptive attributes
5. ANALYTICAL USAGE
Helps business users to:
Analyze from multiple perspectives (View by Time, Product, Location, Customer, etc.)
Perform complex calculations and aggregations (Sum, Avg, Count, Min, Max)
Identify trends, patterns and exceptions
Support strategic decision-making with historical data
Common Use Cases
Sales Analysis
Financial Reporting
Inventory Analysis
Customer Segmentation
Performance Tracking
SUMMARY
✓ Data Cubes organize data in multiple dimensions.
✓ OLAP operations allow users to view data from different angles.
✓ Star and Snowflake schemas model data efficiently in the warehouse.
✓ Fact tables store measurable data and dimension tables provide context.
✓ Together, they enable powerful analysis and informed decisions.
KEY TERMS
• Dimension: Descriptive attributes (Who, What, When, Where)
• Measure: Numeric data for calculation
• Fact Table: Stores measures and foreign keys
• Dimension Table: Stores descriptive attributes
• OLAP: Online Analytical Processing